Xinyi Huang

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EDUCATION

The University of Hong Kong Master's in Electronic Engineering 2025.09-2026.11

- Course: Integrated Circuits, Chip & Memory Design, AI and Edge Processors, Data Compression

Beijing Jiaotong University B.Eng. in Communication Engineering

- GPA: 3.26/4.0 (Top 17%); Third-Class Academic Excellence Scholarship 2021.09-2025.06
- Course: Digital Circuits, Analog Circuits, Digital Signal Processing, Microprocessor Principles & Interfacing

INTERNSHIP EXPERIENCE

Bitmain 2026.06-Present

Objective: Reduce dependency stalls from long-latency FP64 DIV/SQRT operations in RandomX by replacing the 13-cycle SQRT / 9-cycle DIV iterative unit with a shared fixed 5-cycle pipeline, then raising frequency from 2.4 to 2.6 GHz.

Key Contributions:

- Unified DIV and SQRT into a shared **5-cycle fixed-latency, II=1 pipeline** using Goldschmidt division, reciprocal-square-root Newton iterations, 57-bit Q2.55 fixed-point arithmetic, and LUT-based initial approximation.
- For the **2.4 GHz** baseline, partitioned logic to a 5-cycle timing budget with at most one 57x57 multiply per critical stage; implemented result correction and IEEE-754 rounding.
- For the **2.6 GHz** target, used Design Compiler timing reports to identify critical paths, restructured the division LUT, and added parallel pre-computation on the C5 rounding path.
- Used DC power reports to remove redundant clock-gating and operand-isolation logic, reducing power.
- Built directed and constrained-random verification; wrote SVA properties and used VC Formal to verify key functional and pipeline-timing properties.

Results:

- Achieved timing closure at **2.4 GHz and 2.6 GHz** in 5 nm / 0.75 V / 85 C with zero setup violations; maintained 5-cycle latency and II=1, with FP64 outputs matching the reference model bit-for-bit. Power increased by only **20%** versus the previous-generation iterative unit.

PROJECT EXPERIENCE

Transformer Attention Accelerator SoC Subsystem Design 2025.10-2026.01

Objective: Build a software-programmable dual-clock Attention SoC accelerator subsystem for Transformer inference, covering CPU configuration, DMA transfers, clock-domain crossing, result write-back, and interrupt notification.

Key Contributions:

- Designed a **500 MHz bus / 1.25 GHz compute** dual-clock architecture: AHB-Lite for CSR/IRQ control, 128-bit AXI4-MM read/write DMA for Q/K/V and result transfers, and multiple Gray-code asynchronous FIFOs for Bus/Core CDC.
- Integrated an 8x8 FP16 systolic array, Online Softmax, and Causal Tile Scheduler; reused the same array for QK^T and PV, supporting N in {32,48,64,128} and d in {32,64}.
- Built functional and CDC verification with Python, VCS, SVA, Functional Coverage, and VC Static CDC.
- Completed hierarchical synthesis and dual-clock pre-layout STA using Yosys, ASAP7 RVT TT, and OpenSTA.

Results:

- Verified the end-to-end **Memory -> DMA -> CDC -> Attention -> DMA -> Memory** path for N32D32, N64D32 random, and N64D32 causal cases; compared 5,120 outputs with **0 mismatches** and achieved **98.96% functional coverage**.
- Debugged AXI Read DMA data loss under FIFO-full backpressure; fixed the ready/valid handshake and achieved zero errors across N32/N64/Causal end-to-end tests.
- Reduced dual-clock SoC CDC Error/Fatal findings from **65 to 0** using VC Static.
- Implemented scheduler-level Causal Tile Skipping, skipping **28 invalid KV tiles** at N=64, d=32.
- In hierarchical pre-layout STA, key compute blocks and bus glue achieved **WNS/TNS = 0** at 1.25 GHz compute / 500 MHz bus.

SKILLS

HDL/Programming: Verilog, SystemVerilog, C++ | **EDA:** VCS, Verdi, DC, VC Static, Vivado, ModelSim, Cocotb | **English:** IELTS 6.5